No.

Date

不定积分

在实际中，经常要研究：已知 $F(x)=f(x)$（$f(x)$ 已知）$\Leftrightarrow f(x)dx=dF(x)$，求 $F(x)$

定义：设 $f(x)$ 在区间 I 上有定义，若存在一个 $F(x)$ 对每一个 $x\in I$，都有 $F'(x)=f(x)$，称 $F(x)$ 为 $f(x)$ 的一个原函数

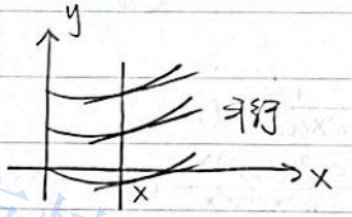
若 $F(x)$ 是 $f(x)$ 在区间 I 上的一个原函数，则 $F(x)+C$ 也是 $f(x)$ 在区间 I 上的原函数 $(C\in R)$

定理：若 $F(x)$ 是 $f(x)$ 在区间 I 上的一个原函数，则 $F(x)+C$（C 常，$C\in R$），是 $f(x)$ 在区间 I 上的全体原函数，称为 $F(x)$ 在 I 上的不定积分，记作 $\int f(x)dx$，即 $F(x)+C=\int f(x)dx$

（$\int$：不定积分号；$f(x)$：被积函数；$f(x)dx$：被积表达式；x：积分变量；C：积分常数）

性质

1. $\left(\int f(x)dx\right)'=\frac{d}{dx}\int f(x)dx=f(x)$
2. $d\int f(x)dx=f(x)dx$
3. $\int f(x)dx=f(x)+C=\int\frac{d}{dx}f(x)dx$
4. $\int df(x)=f(x)+C$



不定积分的几何意义

设 $F(x)$ 是 $f(x)$ 的原函数，则 $\int f(x)dx=F(x)+C$，又 $\left(\int f(x)dx\right)'=(F(x)+C)'=f(x)$

即曲线在相同的 x 处的切线相互平行

双曲余弦 $chx=\frac{e^x+e^{-x}}{2}$

双曲正弦 $shx=\frac{e^x-e^{-x}}{2}$

基本不定积分公式（定义域要相同）

1. $\int 0dx=C$
2. $\int 1dx=\int dx=x+C$
3. $\int x^a dx=\frac{1}{a+1}\cdot x^{a+1}+C\quad(a\neq -1)$
4. $\int x^{-1}dx=\ln|x|+C$
5. $\int a^x dx=\frac{a^x}{\ln a}+C$
6. $\int e^x dx=e^x+C$
7. $\int\cos x dx=\sin x+C$
8. $\int\sin x dx=-\cos x+C$
9. $\int\sec^2 x=\tan x+C$
10. $\int\csc^2 x=-\cot x+C$
11. $\int\sec x\cdot\tan x=\sec x+C$
12. $\int\csc x\cdot\cot x=-\csc x+C$
13. $\int\frac{1}{1+x^2}dx=\arctan x+C=-\operatorname{arccot}x+C_1$
14. $\int\frac{1}{\sqrt{1-x^2}}dx=\arcsin x+C=-\arccos x+C$
15. $\int shx dx=chx+C$
16. $\int chx dx=shx+C$
17. $\int\tan x dx=-\ln|\cos x|+C$
18. $\int\cot x dx=\ln|\sin x|+C$

$ch^2-sh^2=1$　$(sh)'=ch$　$(ch)'=sh$

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不定积分的四大法宝

一. 线性运算法则

若 $\int f(x)dx$, $\int g(x)dx$ 均存在, 且常数 α, β (α, β 不同时为0), 则 $\int [\alpha f(x)+\beta g(x)]dx$ (存在)

$= \alpha\int f(x)dx + \beta\int g(x)dx$

注: $[\alpha\int f(x)dx + \beta\int g(x)dx]' = \alpha[\int f(x)dx]' + \beta[\int g(x)dx]' = \alpha f(x)+\beta g(x)$

$\alpha\int f(x)dx + \beta\int g(x)dx$ 是 $\alpha f(x)+\beta g(x)$ 的原函数, 且含有加C项

eg: $\int \sqrt{x\sqrt{x\sqrt{x}}}\,dx$

$= \int x^{\frac{7}{8}}dx = \frac{8}{15}x^{\frac{15}{8}} + C$

eg: $\int \tan^2 x\,dx$ → 可有 $\tan^2 x = \sec^2 x - 1$

$= \int(\sec^2 x - 1)dx = \int \sec^2 x\,dx - \int 1\,dx = \tan x - x + C$

eg: $\int \frac{1}{\sin^2 x\cos^2 x}dx$

$= \int \frac{\sin^2 x + \cos^2 x}{\sin^2 x\cos^2 x}dx = \int(\sec^2 x + \csc^2 x)dx = \tan x - \cot x + C$

eg: $\int \frac{x^4}{x^2+1}dx$

$= \int \frac{x^4-1+1}{x^2+1}dx = \int(x^2 - 1 + \frac{1}{1+x^2})dx = \frac{1}{3}x^3 - x + \arctan x + C$

二. 不定积分的凑微分法(第一换元法)

若 $F'(u) = f(u)$, 则 $[F(\varphi(x))]' = F'(\varphi(x))\cdot\varphi'(x) = f(\varphi(x))\cdot\varphi'(x)$

如果 $g(x)dx = f(\varphi(x))\varphi'(x)dx$

我们找 $F(u)$, $F'(u) = f(u)$: $f(\varphi(x))d(\varphi(x)) \xrightarrow{\varphi(x)=u} f(u)du = dF(u) = dF(\varphi(x))$

则 $\int g(x)dx = F(\varphi(x)) + C$

或: $\int g(x)dx$ ← $\int f(\varphi(x))\varphi'(x)dx = \int f(\varphi(x))\cdot d(\varphi(x)) \xrightarrow{\varphi(x)=u} \int f(u)du \xrightarrow{F'(u)=f(u)} F(u)+C$

$= F(\varphi(x)) + C$

eg: $\int e^{2x}dx = \int e^{2x}\cdot(2x)'\cdot\frac{1}{2}dx = \frac{1}{2}\int e^{2x}(2x)'dx = \frac{1}{2}\int e^{2x}d(2x) = \frac{1}{2}e^{2x} + C$

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No.

Date

eg. $\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx = \int -\frac{1}{\cos x}\,d(\cos x) = -\ln|\cos x| + C$ ⑰

$\int \cot x\,dx = \int \frac{\cos x}{\sin x}\,dx = \int \frac{1}{\sin x}\,d(\sin x) = \ln|\sin x| + C$ ⑱

凑微-坐微为关系式

$dx = 1\cdot dx = d\frac{1}{a}(ax+b)$ $(a\neq 0)$ 即 $dx = \frac{1}{a}d(ax+b)$

$x\,dx = \frac{1}{2}d(x^2 \pm a^2) = -\frac{1}{2}d(a^2 - x^2)$

$\cos x\,dx = d\sin x$ $\sin x\,dx = -d\cos x$

$\frac{1}{x}dx = d\ln|x| \overset{x>0}{=\!=} d\ln x$ $e^x dx = de^x$

eg. $\int \frac{1}{a^2+x^2}\,dx\ (a\neq 0)$

$= \frac{1}{a^2}\int \frac{1}{1+(\frac{x}{a})^2}\,dx = \frac{1}{a}\int \frac{1}{1+(\frac{x}{a})^2}\,d(\frac{x}{a}) = \frac{1}{a}\arctan\frac{x}{a} + C$ ⑲

eg. $\int \frac{1}{\sqrt{a^2-x^2}}\,dx\ (a>0)$

$= \int \frac{1}{a}\cdot\frac{1}{\sqrt{1-\frac{x^2}{a^2}}}\,dx = \int \frac{1}{\sqrt{1-(\frac{x}{a})^2}}\,d\frac{x}{a} = \arcsin\frac{x}{a} + C$ ⑳

eg. $\int \frac{1}{a^2-x^2}\,dx\ (a\neq 0)$

$= \int \frac{1}{(a-x)(a+x)}\,dx = -\frac{1}{2a}\int (\frac{1}{x-a} - \frac{1}{x+a})\,dx = -\frac{1}{2a}[\int \frac{1}{x-a}\,dx - \int \frac{1}{x+a}\,dx] = -\frac{1}{2a}(\ln|x-a| - \ln|x+a|) + C$ ㉑

$= \frac{1}{2a}\ln|\frac{a+x}{a-x}| + C$

eg. $\int \sec x\,dx = \int \frac{d(x+\frac{\pi}{2})}{\sin(x+\frac{\pi}{2})}$

$= \int \frac{1}{\cos x}\,dx = \int \frac{\cos x}{\cos^2 x}\,dx = \int \frac{\cos x}{1-\sin^2 x}\,dx = \int \frac{1}{1-\sin^2 x}\,d(\sin x) = \frac{1}{2}\ln|\frac{1+\sin x}{1-\sin x}| + C$

$= \frac{1}{2}\ln|\frac{(1+\sin x)^2}{\cos^2 x}| + C = \ln|\sec x + \tan x| + C$ $= \ln|\frac{1+\tan\frac{x}{2}}{1-\tan\frac{x}{2}}|$ ㉒

法二) $= \int \frac{\sec x(\sec x+\tan x)}{\sec x+\tan x}\,dx = \int \frac{1}{\sec x+\tan x}\,d(\sec x+\tan x) = \ln|\sec x+\tan x| + C$

同法: $\int \csc x\,dx = \ln|\csc x - \cot x| + C$ ㉓

$= \ln|\tan\frac{x}{2}|$

eg. $\int e^{ax}dx\ (a\neq 0) = \frac{1}{a}\int a\cdot e^{ax}dx = \frac{1}{a}\int e^{ax}d(ax) = \frac{1}{a}e^{ax} + C$ ㉔

eg. $\int \cos ax\,dx\ (a\neq 0) = \frac{1}{a}\sin ax + C$ $\int \sin ax\,dx = -\frac{1}{a}\cos ax + C$

㉕ ㉖

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三. 不定积分的第二类换元法(等=换元法)

求 $\int f(x)dx$ $f(x)dx = d?$

由 $f(x)dx \xrightarrow{x=\varphi(t)\text{可导}} f(\varphi(t))\varphi'(t)dt \xrightarrow{F'(t)=f(\varphi(t))\cdot\varphi'(t)} dF(t) \xrightarrow{x=\varphi(t)\text{单调}\ t=\varphi^{-1}(x)} dF(\varphi^{-1}(x))$

$\therefore \int f(x)dx = F(\varphi^{-1}(x))+C$

或: $\int f(x)dx \xrightarrow[\text{导,等}]{x=\varphi(t)\text{可}} \int f(\varphi(t))\varphi'(t)dt \xrightarrow{F(t)=f(\varphi(t))\cdot\varphi'(t)} F(t)+C = F(\varphi^{-1}(x))+C$

如果被积函数中含有下列根式,且不能用前面方法,此时,用第二类换元

$\sqrt{a^2-x^2}$, 令 $x=a\sin t$, $t\in[-\frac{\pi}{2},\frac{\pi}{2}]$

$\sqrt{a^2+x^2}$, 令 $x=a\tan t$, $t\in[-\frac{\pi}{2},\frac{\pi}{2}]$ (去根号)

$\sqrt{x^2-a^2}$, 令 $x=a\sec t$, $t\in[0,\frac{\pi}{2})\cup(\frac{\pi}{2},\pi]$ 不仅仅适用于根式

$\sqrt[n]{\frac{ax+b}{cx+d}}$, 令 $\sqrt[n]{\frac{ax+b}{cx+d}}=t$, 解出 $x=\varphi(t)$

① 含多种根号

② 形如 $f(x)^m g(x)^n$

含其中 $f(x)$ $g(x)$ 为一(零点)

哪个简,哪个设为t

eg. $\int\sqrt{a^2-x^2}dx\ (a>0)$

$\xrightarrow{x=\sin t\cdot a}\int\sqrt{a^2-a^2\sin^2 t}\,a\cos t\,dt = a^2\int\cos^2 t\,dt = \frac{a^2}{2}\int(1+\cos 2t)dt$

$= \frac{a^2}{2}(t+\frac{1}{2}\sin 2t)+C = \frac{a^2}{2}(\arcsin\frac{x}{a}+\frac{x}{a^2}\sqrt{a^2-x^2})+C$

eg. $\int\frac{1}{\sqrt{x^2+a^2}}dx\ (a>0)$

$\xrightarrow{x=a\tan t}\int\frac{1}{a|\sec t|}\cdot a\cdot\sec^2 t\,dt = \int\sec t\,dt = \ln|\sec t+\tan t|+C$ (27)

$= \ln\left|\frac{\sqrt{a^2+x^2}}{a}+\frac{x}{a}\right|+C = \ln|x+\sqrt{x^2+a^2}|-\ln|a|+C = \ln(x+\sqrt{x^2+a^2})+C_1$ (令 $C_1=C-\ln|a|$)

$x+\sqrt{x^2+a^2}>0$ 恒成立 $|\sqrt{x^2+a^2}|>|x|$

同理: $\int\frac{1}{\sqrt{x^2-a^2}}dx = \ln|x+\sqrt{x^2-a^2}|+C$

eg. $\int\frac{1}{\sqrt{x}+\sqrt[3]{x}}dx$

$\xrightarrow[x=t^6]{\sqrt[6]{x}=t} 6\int\frac{t^3+1-1}{t+1}dt = 6\int(t^2-t+1-\frac{1}{t+1})dt = 6(\frac{1}{3}t^3-\frac{1}{2}t^2+t-\ln(1+t))+C$

$= 2\sqrt{x}-3\sqrt[3]{x}+6\sqrt[6]{x}-6\ln(\sqrt[6]{x}+1)+C$

eg. $\int x^2(1-x)^{1000}dx$

$\xrightarrow[x=1-t]{t=1-x} -\int(1-t)^2 t^{1000}dt = -\int(1-2t+t^2)t^{1000}dt = \cdots$ (展开)

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eg: $\int \frac{x^2}{(2x+1)^{10}}dx$

$\frac{t=2x+1}{x=\frac{t-1}{2}}$ $\int \frac{t^2+1-2t}{8t^{10}}dt = \frac{1}{8}\int(t^{-8}+t^{-10}-2t^{-9})dt = \cdots$

四. 分部积分

定理: 若$u(x), v(x)$可导，则$\int udv = uv - \int vdu$ 称为分部积分

证: 由$(uv)' = u'v + uv' \Rightarrow uv' = (uv)' - u'v$

$\Rightarrow \int uv'dx = \int (uv)'dx - \int (u'v)dx$

$\Rightarrow \int udv = uv - \int vdu$

证法二: $d(uv) = du\cdot v + udv \Rightarrow udv = d(uv) - vdu$

$\Rightarrow \int udv = \int d(uv) - \int vdu = uv - \int vdu$

求$\int f(x)dx = \int u(x)v'(x)dx = \int udv = uv - \int vdu$

eg. $\int \underset{u}{x}\underset{v'}{e^x}dx$ 令$u=x, v'=e^x \Rightarrow v=e^x$

$= \int x de^x = xe^x - \int e^x dx = xe^x - e^x + C = e^x(x-1)+C$

常见类型:

设$P_k(x)$是x的k次多项式，$\alpha \neq 0$常数

1. $\int \underset{u}{P_k(x)}\underset{v'}{e^{ax}}dx = \int \underset{u}{P_k(x)}\underset{dv}{d(\frac{1}{a}e^{ax})}$ 需要n次分部积分

2. $\int P_k(x)\cos ax dx = \int P_k(x)d(\frac{1}{a}\sin ax)$

3. $\int P_k(x)\sin ax dx = \int P_k(x)d(\frac{-1}{a}\cos ax)$

设$P(x)$为x的函数

4. $\int P(x)f(\arcsin x)dx$ 能凑则凑，若不能凑，令$f(\arcsin x) = u$，$P(x) = v'$，求出v

$f(\arccos x)$ $f(\arctan x)$ $f(\operatorname{arccot} x)$

5. $\int P(x)f(\ln x)dx$ 同上 令$f(\ln x) = u$

eg. $\int (1+x^2)\cos 2x dx = \int (1+x^2)d(\frac{1}{2}\sin 2x) = \frac{1}{2}(1+x^2)\sin 2x - \frac{1}{2}\int \sin 2x d(1+x^2)$

$= \frac{1}{2}(1+x^2)\sin 2x - \underset{u}{\int x}\underset{v'}{\sin 2x}dx = \frac{1}{2}(1+x^2)\sin 2x + \int x d(\frac{1}{2}\cos 2x) = \frac{1}{2}(1+x^2)\sin 2x + \frac{1}{2}x\cos 2x$

$-\frac{1}{2}\int \cos 2x dx = \frac{1}{2}(1+x^2)\sin 2x + \frac{1}{2}x\cos 2x - \frac{1}{4}\sin 2x + C$

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eg. $\int \frac{\arctan x}{(1+x^2)} dx$

$= \int \arctan x \, d(\arctan x) = \frac{1}{2}(\arctan x)^2 + C$

eg. $\int \underset{u}{\arctan x} \underset{dv}{dx} = \arctan x \cdot x - \int x \cdot \frac{1}{1+x^2} dx = x\arctan x - \frac{1}{2}\ln(1+x^2) + C$

eg. $\int \frac{\arctan x}{x^2(1+x^2)} dx$ 线性代换

$= \int (\frac{1}{x^2} - \frac{1}{1+x^2}) \arctan x \, dx = -\frac{1}{x}(\arctan x)^2 + \int \arctan x \, d(-\frac{1}{x})$

$= -\frac{1}{x}(\arctan x)^2 + \arctan x \cdot (-\frac{1}{x}) - \int (-\frac{1}{x}) \cdot \frac{1}{1+x^2} dx$

由 $\int \frac{1}{x} \cdot \frac{1}{1+x^2} = \int (\frac{1}{x} - \frac{x}{1+x^2}) dx = \ln|x| - \frac{1}{2}\ln(1+x^2) + C$

原式 $= -\frac{1}{x}(\arctan x)^2 - \arctan x \cdot \frac{1}{x} + \ln|x| - \frac{1}{2}\ln(1+x^2) + C$

eg. $\int \frac{\ln^2 x}{x} dx = \int \ln^2 x \, d\ln x = \frac{1}{3}(\ln x)^3 + C$

eg. $\int \underset{v'}{\sqrt{x}} \underset{u}{\ln x} dx = \int \ln x \, d(\frac{2}{3} \cdot x^{\frac{3}{2}}) = \frac{2}{3}\ln x \cdot x^{\frac{3}{2}} - \int \frac{2}{3} \cdot x^{\frac{3}{2}} \cdot \frac{1}{x} dx = \frac{2}{3}\ln x \cdot x^{\frac{3}{2}} - \frac{2}{3}\int x^{\frac{1}{2}} dx$

$= \frac{2}{3}\ln x \cdot x^{\frac{3}{2}} - \frac{4}{9}x^{\frac{3}{2}} + C$

eg. 求 $\int \underset{v'}{e^{ax}} \underset{u}{\cos bx} dx$ ($a \neq 0, b \neq 0$ 常) $e^x, \cos x (\sin x)$ 性质差不多

$= \int \cos bx \, d(\frac{1}{a}e^{ax}) = \frac{1}{a}e^{ax}\cos bx + \frac{b}{a}\int \underset{v'}{e^{ax}} \underset{u}{\sin bx} dx$

$= \frac{1}{a}e^{ax}\cos bx + \frac{b}{a}\int \sin bx \, d(\frac{1}{a}e^{ax}) = \frac{1}{a}e^{ax}\cos bx + \frac{b}{a}[\frac{1}{a}e^{ax}\sin bx - \frac{b}{a}\boxed{\int \cos bx \, e^{ax} dx}]$

设 $I = \int e^{ax}\cos bx dx$, 有 $I = \frac{1}{a}e^{ax}\cos bx + \frac{b}{a^2}e^{ax}\sin bx - \frac{b^2}{a^2}I$

$I = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx) + C$

同理可求 $\int e^{ax}\sin bx dx$ } 解方程的思想

eg. $\int \underset{u}{\sqrt{a^2-x^2}} \underset{dv}{dx}$ ($a>0$)

$= \sqrt{a^2-x^2}x - \int x \, d\sqrt{a^2-x^2} = \sqrt{a^2-x^2} \cdot x - \int \frac{-2x^2}{2\sqrt{a^2-x^2}} dx = x\sqrt{a^2-x^2} - \int \frac{a^2-x^2-a^2}{\sqrt{a^2-x^2}} dx$

$= x\sqrt{a^2-x^2} - \boxed{\int \sqrt{a^2-x^2} dx} + \arcsin\frac{x}{a}$

$\therefore \int \sqrt{a^2-x^2} dx = \frac{1}{2}x\sqrt{a^2-x^2} + \frac{1}{2}\arcsin\frac{x}{a} + C$

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Date

换元积分法

第一类换元法

$\int f(e^x)e^x dx = \int f(e^x)\,d(e^x) = F(e^x)+C$　　$\int f(\sin x)\cos x dx = \int f(\sin x)\,d(\sin x) = F(\sin x)+C$　　$\int f(x^{n+1})x^n dx = \frac{1}{n+1}\int f(x^{n+1})\,d(x^{n+1}) = \frac{1}{n+1}F(x^{n+1})+C$

$\int \frac{1}{x}f(\ln x)dx = \int f(\ln x)\,d(\ln x) = F(\ln x)+C$　　$\int f(\tan x)\sec^2 x dx = \int f(\tan x)\,d(\tan x) = F(\tan x)+C$

eg. 求 $\int \frac{1}{x\ln x \ln\ln x}dx$

$= \int \frac{1}{\ln x \cdot \ln\ln x}d\ln x = \int \frac{1}{\ln\ln x}d\ln\ln x = \ln|\ln\ln x|+C$

eg. 求 $\int x^2\sqrt[3]{x^3+2}\,dx$

$= \int \frac{1}{3}\sqrt[3]{x^3+2}\,d(x^3+2) = \frac{1}{3}\times\frac{3}{4}(x^3+2)^{\frac{4}{3}}+C = \frac{1}{4}(x^3+2)^{\frac{4}{3}}+C$

eg. 求 $\int \frac{1}{x^2}e^{\frac{1}{x}}dx$

$= -\int e^{\frac{1}{x}}d\frac{1}{x} = -e^{\frac{1}{x}}+C$

eg. 求 (1) $\int \tan x dx$　(2) $\int \cot x dx$

(1) $= \int \frac{\sin x}{\cos x}dx = -\int \frac{1}{\cos x}d\cos x = -\ln|\cos x|+C$

(2) $= \int \frac{\cos x}{\sin x}dx = \int \frac{1}{\sin x}d\sin x = \ln|\sin x|+C$

eg. 求 (1) $\int \frac{\arctan\sqrt{x}}{\sqrt{x}(1+x)}dx$

$= 2\int \frac{\arctan\sqrt{x}}{1+x}d\sqrt{x} = 2\int \arctan\sqrt{x}\,d(\arctan\sqrt{x}) = (\arctan\sqrt{x})^2+C$

eg. 求 (1) $\int \sin^3 x dx$　(2) $\int \sin^2 x\cos^5 x dx$　(3) $\int \cos^4 x dx$　(4) $\int \sec^6 x dx$

(1) $= \sin x\cdot\sin^2 x = \int(1-\cos^2 x)\cdot\sin x dx = -\int(1-\cos^2 x)\,d\cos x$

(2) $= \int \sin^2 x(1-\sin^2 x)^2\cos x dx = \int \sin^2 x(1-\sin^2 x)^2 d(\sin x)$

(3) $= \int(\frac{1+\cos 2x}{2})^2 dx = \int(\frac{1}{4}+\cos 2x+\frac{1}{4}\cdot\frac{1+\cos 4x}{2})dx$

(4) $= \int \sec^4 x\cdot\sec^2 x dx = \int(1+\tan^2 x)^2 d(\tan x)$

总结：含有三角函数的不定积分问题

$\sin x, \cos x$ 的次数为奇数

$\Rightarrow \sin^m x = \sin x\cdot\sin^{m-1}x = -(1-\cos^2 x)^{\frac{m-1}{2}}d(\cos x)$

$\Rightarrow \cos^m x = \cos x\cdot\cos^{m-1}x = (1-\sin^2 x)^{\frac{m-1}{2}}d(\sin x)$

$\Rightarrow \sin^m x\cos^n x$ (m,n 中至少一个为奇数)

$\sin x, \cos x$ 的次数为偶数

$\Rightarrow$ 降幂公式 $\cos 2x = 2\cos^2 x-1 = 1-2\sin^2 x$ 降幂　目的：将次数变为1次 ($\sin nx, \cos nx$)

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No.

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secx的偶数次方

$\Rightarrow \sec x^m = \sec^2(1+\tan^2 x)^{\frac{m-2}{2}} = \int(1+\tan^2 x)^{\frac{m-2}{2}} d(\tan x)$

eg. 求 $\int \cos 3x \cos 7x\,dx$

$= \frac{1}{2}\int(\cos 5x + \cos 5x)\,dx = \frac{1}{2}(\sin x + \sin 5x) + C$

$\sin mx\cos nx$ / $\sin mx \sin nx$ / $\cos mx \cos nx$

$\Rightarrow$ 积化和差

eg. 求 (1) $\int \frac{x^2+1}{x^4+1}dx$ (2) $\int \frac{1}{x^4+1}dx$

$= \int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx = \int \frac{d(x-\frac{1}{x})}{(x-\frac{1}{x})^2+2} = \frac{1}{\sqrt{2}}\arctan\frac{x-\frac{1}{x}}{\sqrt{2}} + C$

(2) $= \int \frac{1}{x^4+1}dx = \frac{1}{2}\int \frac{1+x^2-x^2+1}{x^4+1}dx = \frac{1}{2}\int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx - \frac{1}{2}\int \frac{x^2+1}{x^4+1}dx = \frac{1}{2}\int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx - \frac{1}{2}\int \frac{1-\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx$

$= \frac{1}{2}\int \frac{d(x-\frac{1}{x})}{(x-\frac{1}{x})^2+2} - \frac{1}{2}\int \frac{d(x+\frac{1}{x})}{(x+\frac{1}{x})^2-2} = \frac{1}{2\sqrt{2}}\arctan\frac{x-\frac{1}{x}}{\sqrt{2}} - \frac{1}{4\sqrt{2}}\ln\left|\frac{x+\frac{1}{x}-\sqrt{2}}{x+\frac{1}{x}+\sqrt{2}}\right| + C$

总结：基本形式 $\frac{x^2+1}{x^4+1}$，$\frac{x^2-1}{x^4+1}$ (利用 $d(x-\frac{1}{x})$, $d(x+\frac{1}{x})$)

$\Rightarrow \int \frac{a_1x^3+a_2x^2+a_3x+a_4}{x^4+1}dx$ $\quad \frac{a_2x^2+a_4}{x^4+1} = a_2\cdot\frac{x^2+1}{x^4+1} + \frac{a_4-a_2}{x^4+1}$ $\quad \frac{a_1x^3+a_3x}{x^4+1} = x\cdot\frac{a_1x^2+a_3}{x^4+1}dx$

$\Rightarrow \int \frac{1}{(x^4+1)^2}dx$ 利用 $\int f(x)dx = f(x)+C$ $\quad = \frac{1}{2}\frac{a_1x^2+a_3}{x^4+1}d(x^2)$ (x^2 换)

设 $\frac{1}{(x^4+1)^2} = \left(\frac{P_{n1}(x)}{x^4+1}\right)' + \frac{Q_{n1}(x)}{x^4+1}$

$\int \frac{1}{(x^4+1)^2}dx = \frac{P_{n1}(x)}{x^4+1} + \int \frac{Q_{n1}(x)}{x^4+1}dx$ 可求

eg. 求 $\int \frac{1+\sin x}{1+\cos x}e^x dx$

法一 $= \int \frac{1+2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}e^x dx = \int(\frac{1}{2}\sec^2\frac{x}{2} + \tan\frac{x}{2})e^x dx = e^x\tan\frac{x}{2} + C$

法二 $= \int(\frac{1}{1+\cos x} + \frac{\sin x}{1+\cos x})e^x dx = \int(u'e^x + ue^x)dx = \int(ue^x)'dx = ue^x + C$

$u = \frac{\sin x}{1+\cos x}$ $\quad u' = \frac{1}{1+\cos x}$

第二类换元法

eg. $\int \frac{1}{x\sqrt{1-x^2}}dx$

$\xrightarrow[t\in(-\frac{\pi}{2},\frac{\pi}{2})]{x=\sin t} \int \frac{\cos t}{\sin t\cos t}dt = \int \csc t\,dt = \ln|\csc t - \cot t| + C = \ln\left|\frac{1}{x} - \frac{\sqrt{1-x^2}}{x}\right| + C$

eg. 求 $\int \frac{1}{x^2\sqrt{x^2+a^2}}dx$

$\xrightarrow[t\in(-\frac{\pi}{2},\frac{\pi}{2})]{x=a\tan t} \frac{1}{a^2}\int \frac{1}{\tan^2 t\sec t}\cdot a\sec^2 t\,dt = \frac{1}{a^2}\int \frac{\cos t}{\sin^2 t}dt = \frac{1}{a^2}\int \frac{1}{\sin^2 t}d(\sin t) = -\frac{1}{a^2}\csc t + C$

$= -\frac{1}{a^2}\frac{\sqrt{a^2+x^2}}{x} + C$

eg. 求 $\int \frac{dx}{ax^n+bx}$

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⇒ 类型: $x^m \cdot f(x^{m+1})\,dx$

$d(t^{m+1})$ $\quad$ $d(x^{m+1})$

No.

Date

令 $x=\frac{1}{t}$ $\int \frac{-\frac{1}{t^2}dt}{a\frac{1}{t^n}+b\frac{1}{t}} = -\int \frac{t^{n-2}}{a+bt^{n+1}}dt = \frac{1}{1-n}\int \frac{dt^{n+1}}{a+bt^{n+1}} = \frac{1}{b(1-n)}\ln|a+bt^{n+1}|+C$

$= \frac{1}{b(1-n)}\ln|a+bx^{\frac{1}{n+1}}|+C$

eg. 求 $\int \frac{1}{x\sqrt{1+x}}dx$

$t=\sqrt{1+x}$ $\int \frac{2t}{(t^2-1)t}dt = \int \frac{2}{t^2-1}dt = \int(\frac{1}{t-1}-\frac{1}{t+1})dt = \ln|\frac{t-1}{t+1}|+C$

$= \ln|\frac{\sqrt{1+x}-1}{\sqrt{1+x}+1}|+C$

形如 $\sqrt{ax+b}$, 整体令 $t=\sqrt{ax+b}$

eg. 求 $\int \frac{1}{x+2+\sqrt{x+1}}dx$

拼凑(吹泡泡)

$t=\sqrt{x+1}$ $\int \frac{2t+1}{t^2+t+1}dt = \int \frac{2t+1}{t^2+t+1}dt - \int \frac{1}{t^2+t+1}dt = \ln(t^2+t+1) - \int \frac{1}{(t+\frac{1}{2})^2+\frac{3}{4}}dt$

$= \ln(t^2+t+1) - \frac{2}{\sqrt{3}}\arctan\frac{2t+1}{\sqrt{3}}+C = \ln(x+2+\sqrt{x+1}) - \frac{2}{\sqrt{3}}\arctan\frac{2\sqrt{x+1}+1}{\sqrt{3}}+C$

eg. 求 (1) $\int \frac{dx}{x^2+x+4}$ (2) $\int \frac{dx}{\sqrt{4x^2+9}}$ (3) $\int \frac{dx}{\sqrt{1+x-x^2}}$ (4) $\int x\sqrt{\frac{x}{2a-x}}dx$ $(a>0)$

$= \int \frac{dx}{(x+1)^2+(\sqrt{3})^2}dx = \frac{1}{\sqrt{3}}\arctan\frac{x+1}{\sqrt{3}}+C$

(2) $= \int \frac{dx}{\sqrt{4x^2+9}} = \frac{1}{2}\int \frac{d(2x)}{\sqrt{(2x)^2+3^2}} = \frac{1}{2}\ln|2x+\sqrt{4x^2+9}|+C$ } 配方+次式

(3) $= \int \frac{d(x-\frac{1}{2})}{\sqrt{(\frac{\sqrt{5}}{2})^2-(x-\frac{1}{2})^2}} = \arcsin\frac{2x-1}{\sqrt{5}}+C$

(4) $x=2a\sin^2t$, $t\in(0,\frac{\pi}{2})$ $\int 2a\sin^2t\sqrt{\frac{2a\sin^2t}{2a-2a\sin^2t}}\,4a\sin t\cos t\,dt = 8a^2\int \sin^4t\,dt$

法一) 标准做法 令 $t=\sqrt{\frac{x}{2a-x}}$

$= \int \frac{2at^3}{t^2+1}\cdot\frac{4at}{(t^2+1)^2}dt = 8a^2\int \frac{t^4}{(t^2+1)^3}dt = 8a^2\int(\frac{t^4-1}{(t^2+1)^3}+\frac{1}{(t^2+1)^3})dt$

$= 8a^2\int(\frac{1}{t^2+1}-\frac{2}{(t^2+1)^2}+\frac{1}{(t^2+1)^3})dt$

$t=\tan u$ $8a^2\int(\cos^2u-2\cos^4u+\cos^6u)\cdot\frac{1}{\cos^2u}du$

$= 8a^2\int(1-2\cos^2u+\cos^4u)du = 8a^2\int(\cos^2u-1)^2du = 8a^2\int \sin^4u\,du$

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No.

Date

必备公式总结：

$\int 0\,dx = C$

$\int 1\,dx = x + C$

$\int x^a dx = \frac{1}{a+1}x^{a+1} + C \quad (a \neq -1)$

$\int x^{-1} dx = \ln|x| + C$

$\int a^x dx = \frac{a^x}{\ln a} + C$

$\int e^x dx = e^x + C$

$\int \cos x\,dx = \sin x + C$

$\int \sin x\,dx = -\cos x + C$

$\int \sec^2 x\,dx = \tan x + C$

$\int \csc^2 x\,dx = -\cot x + C$

$\int \sec x \cdot \tan x = \sec x + C$

$\int \csc x \cdot \cot x = -\csc x + C$

$\int \frac{1}{1+x^2} dx = \arctan x + C = -\operatorname{arccot} x + C$

$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C = -\arccos x + C$

$\int \operatorname{sh} x\,dx = \operatorname{ch} x + C$

$\int \operatorname{ch} x\,dx = \operatorname{sh} x + C$

$\int \tan x\,dx = -\ln|\cos x| + C$

$\int \cot x\,dx = \ln|\sin x| + C$

$\int \frac{1}{a^2+x^2} dx = \frac{1}{a}\arctan\frac{x}{a} + C$

$\int \frac{1}{a^2-x^2} dx = \frac{1}{2a}\ln\left|\frac{x+a}{x-a}\right| + C$

$\int \frac{1}{\sqrt{x^2+a^2}} dx = \ln\left|x + \sqrt{a^2+x^2}\right| + C$

$\int \frac{1}{\sqrt{x^2-a^2}} dx = \ln\left|x + \sqrt{x^2-a^2}\right| + C$

$\int \frac{1}{\sqrt{a^2-x^2}} dx = \arcsin\frac{x}{a} + C$

$\int \sec x\,dx = \ln|\sec x + \tan x| + C$

$\int \csc x\,dx = \ln|\csc x - \cot x| + C$

$\int e^{ax} dx = \frac{1}{a}e^{ax} + C$

$\int \cos ax\,dx = \frac{1}{a}\sin ax + C$

$\int \sin ax\,dx = -\frac{1}{a}\cos ax + C$

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No.

Date

分部积分

原则：(1) 由v'易求出v(凑微法得到)　(2)一般$\int u'v\,dx$比$\int uv'\,dx$好求

LIATE法 ⇒ 选择作为u的优先顺序

$$\begin{cases} \text{对数} \\ \text{反三角函数} \end{cases} \rightarrow \begin{cases} \text{代数} \\ \text{幂函数} \end{cases} \rightarrow \begin{cases} \text{三角函数} \\ \text{指数函数} \end{cases}$$

eg. 求$\int P_n(x)e^{ax}dx$形式的不定积分

法一) 分部

法二) 待定系数法

设$\int P_n(x)e^{ax}dx = Q_n(x)e^{ax}+C$，其中$Q_n(x)$为待定系数的与$P_n(x)$同次多项式

$\int P_n(x)e^{ax}dx = Q_n(x)e^{ax}+C$

两边求导：$P_n(x)e^{ax} = Q_n'(x)e^{ax} + aQ_n(x)e^{ax}$

$\Rightarrow P_n(x) = Q_n'(x) + aQ_n(x)$

eg. 求$\int \arctan x \cdot x\,dx$

$= \frac{1}{2}\int \arctan x\, d(x^2+1) = \frac{1}{2}\arctan x\cdot(x^2+1) - \int (x^2+1)\cdot\frac{1}{1+x^2}dx$

$= \frac{1}{2}\arctan x(x^2+1) - x + C$

eg. 求$\int e^x \sin x\,dx$

$= \int \sin x\, de^x = e^x\sin x - \int e^x\cos x\,dx = e^x\sin x - \int \cos x\, de^x = e^x\sin x - e^x\cos x - \int e^x\sin x\,dx$

移项：$\int e^x\sin x\,dx = \frac{1}{2}e^x(\sin x - \cos x) + C$　　移项解方程法(三角函数，指数相关)

eg. 求$\int \sec^3 x\,dx$

$= \sec x\cdot\tan x - \int \tan^2 x\cdot\sec x\,dx = \sec x\cdot\tan x - \int(\sec^2 x - 1)\sec x\,dx$

$\Rightarrow 2\int \sec^3 x\,dx = \sec x\tan x + \int \sec x\,dx$

$\Rightarrow I_n = \int \sec^n x\,dx$

$I_0 = x + C$　$I_1 = \ln|\sec x + \tan x| + C$　$I_2 = \tan x + C$

$n \geq 3$, $I_n = \int \sec^{n-2}x\cdot\sec^2 x\,dx = \sec^{n-2}x\cdot\tan x - \int (n-2)\sec^{n-2}x(\sec^2 x - 1)\,dx$

$= \sec^{n-2}x\tan x - (n-2)\int(\sec^n x - \sec^{n-2}x)\,dx$

$I_n = \frac{1}{n-1}\sec^{n-2}x\tan x + \frac{n-2}{n-1}I_{n-2}$　递推公式

递推公式法：还可以求$I_n = \int \sin^n x\,dx$，$I_n = \int \cos^n x\,dx$

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No.

Date

根号 → 意识代换

eg. 求 $\int e^{\sqrt{x}}dx$

$\xrightarrow{t=\sqrt{x}}$ $\int e^t \cdot 2t\,dt$ (分部)

eg. 求 $\int \ln^2(x+\sqrt{1+x^2})dx$ ($u=\ln^2(x+\sqrt{1+x^2})$)

$= x\ln^2(x+\sqrt{1+x^2})dx - 2\int \frac{x}{\sqrt{1+x^2}}\ln(x+\sqrt{1+x^2})dx = x\ln^2(x+\sqrt{1+x^2}) - 2\sqrt{1+x^2}\ln(x+\sqrt{1+x^2}) + 2x + C$

$\ln^2(x+\sqrt{1+x^2})\cdot x -$

$2\int \frac{x}{\sqrt{1+x^2}}\ln(x+\sqrt{1+x^2})dx$ 全部换为

$x\ln^2(x+\sqrt{1+x^2})dx - \int 2\ln(x+\sqrt{1+x^2})\cdot\frac{x}{\sqrt{1+x^2}}\cdot\frac{1}{2}d(x^2+1)$

分析条件：出现含有复合关系根号

即令 $x=\tan u$，$\frac{1+\frac{x}{\sqrt{1+x^2}}}{x+\sqrt{1+x^2}}$

但是积分完会多次，非常复杂，$=\frac{1}{\sqrt{1+x^2}}$ $2\int \ln(x+\sqrt{1+x^2})\,d(x^2+1)^u$

那怎么办呢？

我们平常经常讲过，当下有的方法 $x^2+1=u^2$

那你不出来时候怎么办非常麻烦 $x^2=u^2-1$

分离后3. $(\sqrt{u^2-1}+u)^{\frac{1}{2}}$

所以一定要放回原题考!!!

$2\sqrt{u}$

$d\sqrt{\quad}$ 令 $x=\tan u$

$= 2\int \ln(\tan u+\sec u)\cdot\frac{\tan u}{\sec u}\cdot\sec^2 u\,du$

另外，可以总结一下，$\ln\frac{\sin x+1}{\cos x}\cdot\frac{\sin x}{\cos^2 x}dx$

出现 arc，ln 必为分部

积为

$2\sqrt{x^2+1}$

$\int \frac{2x}{\sqrt{1+x^2}}\ln(x+\sqrt{1+x^2})dx$

v' u

$\ln(x+\sqrt{1+x^2})\cdot 2\sqrt{x^2+1} - \int 2\sqrt{x^2+1}\cdot\frac{1}{\sqrt{1+x^2}}dx$

$= \quad +2x$

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No.

Date

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eg. 求 $\int e^{\sqrt{x}}dx$

$\underline{t=\sqrt{x}}$ $\int e^t \cdot 2t\,dt$ (分部)

eg. 求 $\int \ln^2(x+\sqrt{1+x^2})dx$ ($u=\ln^2(x+\sqrt{1+x^2})$)

$= x\ln^2(x+\sqrt{1+x^2})dx - 2\int \frac{x}{\sqrt{1+x^2}}\ln(x+\sqrt{1+x^2})dx = x\ln^2(x+\sqrt{1+x^2}) - 2\sqrt{1+x^2}\ln(x+\sqrt{1+x^2}) + 2x + C$

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有理函数，三角函数的有理式，无理函数的不定积分

一. 有理函数的不定积分

定义：设 $P_n(x)$，$Q_m(x)$ 分别是 n 次、m 次多项式，称 $\frac{P_n(x)}{Q_m(x)}$ 为有理函数

当 $n \geq m$，称为有理假分式函数

当 $n < m$，称为有理真分式函数

⇒ 利用多项式除法一定可以把有理假分式化成多项式与有理真分式之和

eg. $\frac{x^4+3x^2+x+1}{x^2+x+1} = x^2-x+3-\frac{x+2}{x^2+x+1}$

$$
\begin{array}{r|l}
x^2+x+1 & x^4+0x^3+3x^2+x+1 \\
 & x^4+x^3+x^2 \\
\hline
 & -x^3+2x^2+x \\
 & -x^3-x^2-x \\
\hline
 & 3x^2+2x+1 \\
 & 3x^2+3x+3 \\
\hline
 & -x-2
\end{array}
$$

缺少的次要补齐

有理真分式函数的化简

要求：将分母在实数范围内分解成一次因式与二次因式的乘积，二次因式无法再分解（$\Delta<0$）

eg. $\frac{x^2-x+5}{(x-1)(x-2)^3(x^2-x+1)(x^2+x+1)^2} = \frac{A}{x-1}+\frac{B}{x-2}+\frac{C}{(x-2)^2}+\frac{D}{(x-2)^3}+\frac{Ex+F}{x^2-x+1}+\frac{Gx+H}{x^2+x+1}+\frac{Ix+J}{(x^2+x+1)^2}$

通分求解未知数

eg. $\frac{x^3+x-1}{(x-1)^3(x^2-x+1)^3} = \frac{A}{x-1}+\frac{B}{(x-1)^2}+\frac{Cx+D}{x^2-x+1}+\frac{Ex+F}{(x^2-x+1)^2}+\frac{Gx+H}{(x^2-x+1)^3}$

技巧：对于一次多项式，无论多少次幂，上面一定是常数

二次多项式，上面一定是一次多项式

且有几次幂就对应有几项 (m)，从 $1 \to m$ 次

⇒ $\int \frac{A}{(x-a)^n}dx$　　$\int \frac{Ax+B}{(x^2+px+q)^n}dx$

第一类最简分式　　第二类最简分式

当 $n=1$ 时，$\int \frac{A}{(x-a)}dx = A\int \frac{1}{x-a}d(x-a) = A\ln|x-a|+C$

当 $n>1$ 时，$\int \frac{A}{(x-a)^n}dx = A\int (x-a)^{-n}dx = \frac{A}{1-n}\frac{1}{(x-a)^{n-1}}+C$

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说明积分一下子求不出来，而且需要用递推公式（很可能用分部积分）

eg. 求 $I_n=\int\frac{1}{(x^2+a^2)^n}dx$ $(n\in N, a\neq 0)$

$I_n=\frac{1}{a^2}\int\frac{a^2+x^2-x^2}{(x^2+a^2)^n}dx$ → 构造 I_{n+1}

$=\frac{1}{a^2}I_{n+1}-\frac{1}{a^2}\int x\cdot\frac{x}{(x^2+a^2)^n}dx$ → 易求原函数 [分部]

令 $u'=\frac{x}{(x^2+a^2)^n}$ $u=\int\frac{x}{(x^2+a^2)^n}dx=\frac{1}{2}\int(x^2+a^2)^{-n}d(x^2+a^2)=-\frac{1}{2(n-1)}\frac{1}{(x^2+a^2)^{n-1}}$

$\Rightarrow I_n=\frac{1}{a^2}I_{n+1}+\frac{1}{2a^2(n-1)}\int x\,d\frac{1}{(x^2+a^2)^{n-1}}$

$=\frac{1}{a^2}I_{n+1}+\frac{1}{2a^2(n-1)}\cdot\frac{x}{(x^2+a^2)^{n-1}}-\frac{1}{2a^2(n-1)}\int\frac{1}{(x^2+a^2)^{n-1}}dx$ I_{n-1}

$=\frac{1}{a^2}\cdot\frac{2n-3}{2(n-1)}I_{n+1}+\frac{1}{2a^2(n-1)}\cdot\frac{x}{(x^2+a^2)^{n-1}}$ (递推公式)

$I_1=\int\frac{1}{x^2+a^2}dx=\frac{1}{a}\arctan\frac{x}{a}+C$

$\Rightarrow\int\frac{Ax+B}{(x^2+px+q)^n}dx$ $(p^2-4q<0)$

若 $A\neq 0$ $\int\frac{Ax+B}{(x^2+px+q)^n}dx=\frac{A}{2}\int\frac{2x+\frac{2B}{A}+p-p}{(x^2+px+q)^n}dx$ ☆

$=\frac{A}{2}\int\frac{2x+p}{(x^2+px+q)^n}dx+(B-\frac{Ap}{2})\int\frac{dx}{(x^2+px+q)^n}$

$=\frac{A}{2}\int\frac{d(x^2+px+q)}{(x^2+px+q)^n}+(B-\frac{Ap}{2})\int\frac{d(x+\frac{p}{2})}{[(x+\frac{p}{2})^2+(\frac{4q-p^2}{2})^2]^n}$

凑方 $\Rightarrow(x^2+a^2)^n$

eg. 求 $\int\frac{1}{x^4-1}dx$

法一 $=\int\frac{1}{2}(\frac{1}{x^2-1}-\frac{1}{x^2+1})dx=\frac{1}{2}\int[\frac{1}{2}(\frac{1}{x-1}-\frac{1}{x+1})-\frac{1}{x^2+1}]dx$

$=\frac{1}{4}\ln|\frac{x-1}{x+1}|-\frac{1}{2}\arctan x+C$

法二 $=\frac{1}{(x+1)(x-1)(x^2+1)}=\frac{A}{x+1}+\frac{B}{x-1}+\frac{Cx+D}{x^2+1}$

用赋值法 $1=A(x-1)(x^2+1)+B(x+1)(x^2+1)+(Cx+D)(x+1)(x-1)$

拆+简单的数 令 $x=-1$ $1=-4A$ $A=-\frac{1}{4}$

令 $x=1$ $1=4B$ $B=\frac{1}{4}$

令 $x=0$ $1=-A+B-D$ $D=-\frac{1}{2}$

令 $x=2$ $1=5A+15B+3(2C+D)$ $C=$

eg. 求 $\int\frac{x^4+2x^3+2x+1}{x^3+x+1}dx$

$=\int(x^2+x-2+\frac{3x+3}{x^3+x+1})dx=\frac{1}{3}x^3+\frac{1}{2}x^2-2x+3\int\frac{x+1}{x^3+x+1}dx$

$\int\frac{x+1}{x^3+x+1}dx=\frac{1}{2}\int(\frac{2x+1}{x^3+x+1}+\frac{1}{x^3+x+1})dx=\frac{1}{2}\ln|(x^3+x+1)|+\frac{1}{2}\int\frac{1}{(x+\frac{1}{2})^2+\frac{3}{4}}dx$

$=\frac{1}{2}\ln|(x^3+x+1)|+\frac{1}{2}\times\frac{2}{\sqrt{3}}\arctan\frac{2(x+\frac{1}{2})}{\sqrt{3}}+C$

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eg: 求 $\int \frac{x^4+1}{x^6+1}dx$ → 因式分解 (a^3+b^3)

$= \int \frac{x^4-x^2+1+x^2}{(x^2+1)(x^4-x^2+1)}dx = \int \frac{1}{x^2+1}dx + \int \frac{x^2}{(x^3)^2+1}dx$

$= \arctan x + \frac{1}{3}\int \frac{d(x^3)}{(x^3)^2+1}$

$= \arctan x + \frac{1}{3}\arctan(x^3) + C$

二. 三角函数有理式的不定积分

定义：由 $\sin x, \cos x, \tan x, \cot x, \sec x, \csc x$ 及常数经过四则运算得到的表达式，记为 $R(\sin x, \cos x, \tan x, \cot x, \sec x, \csc x)$ 称为三角函数的有理式

$= R(\sin x, \cos x)$

⇒ 求 $\int R(\sin x, \cos x)dx$ ⇒ 万能公式 (令 $t = \tan\frac{x}{2}$)

$\cos x = 2\cos^2\frac{x}{2} - 1 = \frac{2}{\sec^2\frac{x}{2}} - 1 = \frac{2}{1+\tan^2\frac{x}{2}} - 1 = \frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}$

$\therefore \cos x = \frac{1-t^2}{1+t^2}$

$\sin x = 2\sin\frac{x}{2}\cos\frac{x}{2} = 2\frac{\sin\frac{x}{2}}{\cos\frac{x}{2}}\cdot\cos^2\frac{x}{2} = \frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}} = \frac{2t}{1+t^2}$

$\begin{cases} \sin x = \frac{2t}{1+t^2} \\ \cos x = \frac{1-t^2}{1+t^2} \end{cases}$ $\tan\frac{x}{2} = t$ $x = 2\arctan t$ $dx = \frac{2}{1+t^2}dt$

$\therefore \int R(\sin x, \cos x)dx = \int R\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right)\frac{2}{1+t^2}dt$ 转化为有理函数的积分

定理：有理函数不定积分一定存在，结果是多项式或有理函数或对数函数或反三角函数

1. $\int R(\sin^2 x, \sin x\cos x, \cos^2 x)dx$

⇒ $\sin^2 x = \frac{\sin^2 x}{\cos^2 x + \sin^2 x}\cdot\cos^2 x = \frac{\tan^2 x}{1+\tan^2 x}$

$\sin x\cos x = \frac{\sin x}{\cos x}\cdot\cos^2 x = \frac{\tan x}{1+\tan^2 x}$

$\cos^2 x = \frac{1}{\sec^2 x} = \frac{1}{1+\tan^2 x}$

原式 $= \int R(\tan x)dx = \int R(\tan x)\cdot\frac{\sec^2 x}{\sec^2 x}dx$ 能凑则凑，如果不能凑，令 $t = \tan x$

$= \int R(\tan x)\frac{1}{1+\tan^2 x}d(\tan x)$ 原式 $= \int R(t)\frac{1}{1+t^2}dt$ (有理函数)

eg: 求 $\int \frac{1}{\sin x\cos^3 x}dx$

$= \int \frac{1}{\sin x\cos x}\cdot\sec^2 x\,dx = \int \frac{1+\tan^2 x}{\tan x}d(\tan x) = \ln|\tan x| + \frac{1}{2}\tan^2 x + C$

法二) $= \int \frac{\sin^2 x + \cos^2 x}{\sin x(\cos x)^3}dx = \int \frac{\sin x}{\cos^3 x}dx + \int \frac{1}{\sin x\cos x}dx$

KOKUYO

No.

Date

成 $\frac{1}{\sin x\cos x}=\frac{\sin^2x+\cos^2x}{\sin x\cos x}=\tan x+\cot x$

$=-\int(\cos x)^{-3}d(\cos x)+\int\frac{2}{\sin 2x}dx=\frac{1}{2}\sec^2x+\int\csc 2x\,d(2x)$

$=\frac{1}{2}\sec^2x+\ln|\csc 2x-\cot 2x|+C$

2. $\int\sin^nx\cdot\cos^mx\,dx$

(1) n, m中至少有1个是奇数，另1个指数可以是任意的偶数，此时把奇数分离出一次幂，用凑微分

eg. $\int\sin^{\frac{1}{3}}x\cos^3x\,dx$

$=\int\sin^{\frac{1}{3}}x\cos^2x\cos x\,dx=\int\sin^{\frac{1}{3}}x(1-\sin^2x)\,d(\sin x)=\frac{3}{5}\sin^{\frac{5}{3}}x-\frac{3}{13}\sin^{\frac{13}{3}}x+C$

(2) m, n均为偶数，利用三角公式降幂，或用线性运算法则或用(1)

eg. $\int\cos^4x\sin^2x\,dx$

$=\int\frac{1}{8}(1+\cos 2x)(1-\cos 2x)\sin^2 2x\,dx=\frac{1}{8}\int\sin^32x\,dx+\frac{1}{8}\int\cos 2x\cdot\sin^22x\,dx$

$=\frac{1}{16}\int(1-\cos 4x)\,dx+\frac{1}{16}\int\cos 2x\cdot\sin^22x\,d(2x)=\frac{1}{16}\int(1-\cos 4x)\,dx+\frac{1}{16}\int\sin^22x\,d(\sin 2x)$

$=\frac{1}{16}(x-\frac{1}{4}\sin 4x)+\frac{1}{16}\times\frac{1}{3}(\sin 2x)^3+C$

3. $\int\sin mx\cdot\cos nx\,dx$

⇒积化和差

eg. $\int\sin 4x\cdot\cos 3x\,dx$

$=\frac{1}{2}\int(\sin 7x+\sin x)\,dx=-\frac{1}{14}\cos 7x-\frac{1}{2}\cos 3x+C$

$\sin\alpha+\sin\beta=2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$

$\sin\alpha-\sin\beta=2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$

$\cos\alpha+\cos\beta=2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$

$\cos\alpha-\cos\beta=-2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$

三. 无理函数的不定积分

1. $\int R(x,\sqrt[n]{\frac{ax+b}{cx+d}})\,dx$

⇒令 $t=\sqrt[n]{\frac{ax+b}{cx+d}}$ ∴ $x=\varphi(t)$有理式

原式 $=\int R(t,\varphi(t))\varphi'(t)\,dt$ 转化成有理函数的不定积分

$\sqrt[n]{ax+b}=t$ $\sqrt{(x-a)(x-b)}=|x-a|\sqrt{\frac{x-b}{x-a}}=(x-a)\sqrt{\frac{x-b}{x-a}}\Rightarrow\sqrt{\frac{x-b}{x-a}}=t$

积分范围并不影响所求得原函数

2. $\int R(x,\sqrt{ax^2+bx+c})\,dx$ $(a\neq 0)$

首先把根式化成下列形式之一：$\sqrt{q^2-\varphi^2(x)}$ 或 $\sqrt{q^2+\varphi^2(x)}$ 或 $\sqrt{\varphi^2(x)-q^2}$

其中 $\varphi(x)=hx+p\,(h\neq 0)$，q为常数 $dx=\frac{1}{h}d(hx+p)=\frac{1}{h}d(\varphi(x))$

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No.

Date

在心中把$\varphi(x)$看成u，能凑则凑，如果不能凑就要去根式

令$\varphi(x)=q\sin t$，$d(\varphi(x))=q\cos t\,dt$，$x=\frac{1}{a}(q\sin t-p)$

令$\varphi(x)=q\tan t$ …

令$\varphi(x)=q\sec t$ …

eg. $\int\frac{1}{\sqrt{x^2+x+1}}dx$ $\int\frac{1}{\sqrt{x^2+a^2}}dx=\ln(x+\sqrt{x^2+a^2})+C$

$=\int\frac{1}{\sqrt{(x+\frac{1}{2})^2+\frac{3}{4}}}dx=\int\frac{1}{\sqrt{(x+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2}}d(x+\frac{1}{2})=\ln(x+\frac{1}{2}+\sqrt{x^2+x+1})+C$

eg. $\int\sqrt{2+x-x^2}\,dx$

$=\int\sqrt{\frac{9}{4}-(x-\frac{1}{2})^2}\,d(x-\frac{1}{2})$

$\frac{x-\frac{1}{2}=\frac{3}{2}\sin t}{t\in(-\frac{\pi}{2},\frac{\pi}{2})}$ $\int\frac{3}{2}\cos t\times\frac{3}{2}\cos t\,dt=\frac{9}{4}\int\frac{1+\cos 2t}{2}\,dt=\frac{9}{4}(\frac{1}{2}t+\frac{1}{4}\sin 2t)+C$

$=\frac{9}{8}\arcsin\frac{2}{3}(x-\frac{1}{2})+\frac{1}{2}(x-\frac{1}{2})\sqrt{2+x-x^2}+C$

eg. $\int e^{|x|}dx$ ⇒分段函数

$e^{|x|}=\begin{cases}e^x, & x\geq 0\\ e^{-x}, & x<0\end{cases}$

当$x<0$时，$\int e^{|x|}dx=\int e^{-x}dx=-e^{-x}+C_1$

当$x\geq 0$时，$\int e^{|x|}dx=\int e^x dx=e^x+C_2$

不可写成C（C_1、C_2不同，且C_1、C_2有关系）

由原函数在原点处可导必连续，显然右连续

已知右连续，即$\lim\limits_{x\to 0^+}(e^x+C_2)=1+C_2=-1+C_1$ ∴$C_2=-2+C_1$

∴$\int e^{|x|}dx=\begin{cases}-e^{-x}+C, & x\leq 0\\ e^x-2+C, & x>0\end{cases}$

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No.

Date

eg. $\int \frac{x}{x-\sqrt{x^2-1}}\,dx$

$= \int \frac{x(x+\sqrt{x^2-1})}{(x-\sqrt{x^2-1})(x+\sqrt{x^2-1})}\,dx = \int (x^2+x\sqrt{x^2-1})\,dx$

eg. $\int \frac{1}{1+\tan x}\,dx$

法一) $= \int \frac{1}{1+\tan x}\cdot\frac{\sec^2 x}{\sec^2 x}\,dx = \int \frac{1}{(1+\tan x)(1+\tan^2 x)}\,d(\tan x)$

令 $t=\tan x$

法二) $\int \frac{1}{1+\tan x}\,dx = \int \frac{\cos x}{\sin x+\cos x}\,dx = \frac{1}{2}\int \frac{\cos x-\sin x+\cos x+\sin x}{\sin x+\cos x}\,dx$

$= \frac{1}{2}(x+\ln|\sin x+\cos x|)+C$

eg. $\int \frac{x^2}{(1-x)^{100}}\,dx$

$\overline{t=1-x}$ $\int \frac{(1-t)^2}{t^{100}}\times(-1)\,dt$

eg. $\int \frac{\ln\tan x}{\sin x\cos x}\,dx$

法一) 令 $\tan x = t$

法二)

eg. $\int \frac{1}{1+\sin x}\,dx$

法一) 万能公式

法二) $= \int \frac{1-\sin x}{\cos^2 x}\,dx$

eg. $\int x^3\sqrt{4-x^2}\,dx$ (x^3: 奇数)

法一) 令 $x=2\sin t$

法二) $= -\frac{1}{2}\int x^2\sqrt{4-x^2}\,d(4-x^2)$ $\overline{t=\sqrt{4-x^2}}$ $-\frac{1}{2}\int (4-t^2)t\,d(t^2)$

Tip: 当所含 x 次数为奇数时，凑成 $\int f(\sqrt{ax+b}, x)\,dx$ 的形式

eg. $\int e^x(x+\ln x)\,dx$

$= \int e^x\cdot x\,dx + \int e^x\cdot\ln x\,dx = e^x\cdot\ln x - \int e^x\ln x\,dx + \int e^x\ln x\,dx + C = e^x\cdot\ln x + C$

eg. $\int \frac{1}{(x+1)\sqrt{x^2+1}}\,dx$

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